



The Impact of the PEAK Curriculum on Standardized Measures of Intelligence: A Systems Level Randomized Control Trial

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Abstract

Objectives The present study examined the effects of the Promoting the Emergence of Advanced Knowledge (PEAK) curriculum on standardized measures of intelligence and PEAK pre-assessment scores participants with autism spectrum disorders (ASD).

Methods Direct support staff at a specialized school for students with special needs implemented teaching programs from either the PEAK curriculum ($n = 26$) or the standard academic curriculum ($n = 26$) using a randomized control design over one semester.

Results Statistically significant change scores were observed in raw IQ on two of three sub-scales, vocabulary and information, and full-scale IQ in a sub-group of participants that scored above the floor of the IQ test at either baseline or following intervention. Statistically significant improvements were also observed in PEAK pre-assessment scores in participants in the PEAK group compared to the control group.

Conclusions Findings indicate that the PEAK curriculum has the potential to positively impact standardized measures of intelligence when implemented for a relatively short period of time by direct support level personnel. These results are especially important given the widespread use of IQ tests to determine placement and eligibility criteria for individuals with special needs.

Keywords Autism spectrum disorder · PEAK · Intelligence · IQ

The Center for Disease Control (Baio et al., 2018) reports US prevalence rates of autism spectrum disorders (ASD) at 1 in 59. The unique educational needs of individuals with ASD have led many parents and school districts to explore education programs that offer autism-specific teaching approaches (Lauderdale-Littin et al., 2013). Interventions based in applied behavior analysis (ABA) are considered best practice for the treatment of ASD (e.g., Heyvaert et al., 2014; Irwin & Axe, 2019; Peters-Scheffer et al., 2011; Rosenwasser & Axelrod, 2001). Several curricula based in behavior analysis have been developed to teach targeted skills to students with ASD, some have which have demonstrated improvements in cognitive performance. Lovaas (1987) produced significant improvements in IQ score following intensive behavioral intervention in 19 participants diagnosed with ASD. These outcomes were

far superior to 40 control participants that did not receive intensive behavioral intervention. Similar findings have been replicated in more recent literature (Cohen et al., 2006; Sallows et al., 2005). Although the results cited above are encouraging, a comprehensive behavioral approach to language development was not evaluated (Hayes et al., 2001).

Relational frame theory (RFT) is a branch of behavioral science that seeks to understand and teach language using the principles of verbal behavior initially outlined by Skinner (1957) and expanded to include more complex relational abilities (Barnes-Holmes et al., 2000; Sidman, 1971). RFT is characterized by the principles of mutual entailment, combinatorial entailment, and the transformation of stimulus functions (Hayes et al., 2001). Mutual entailment refers to the derived bidirectionality of relations (Hayes et al., 2001). Combinatorial entailment refers to the process by which two or more mutually entailed relations combine (Hayes et al., 2001). If a verbally competent human learns the relationship between stimulus A and stimulus B, and B to C, they will now show mutual entailment (relate B to A, C to B) and combinatorial entailment (A to C) without direct training. The

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